



Paper ID : 261571

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Subject Code: BCE601

Roll No:

BTECH

(SEM. VI) THEORY EXAMINATION 2025-26

DESIGN OF CONCRETE STRUCTURES

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably. Use 20 & Fe 415 if not given.

SECTION A

1. Attempt all questions in brief.

02 x 7 = 14

Q no.	Question	CO	Level
a.	Define M25.	1	K2
b.	Sketch continuous slab.	3	K2
c.	Define meaning of doubly reinforced beams.	2	K2
d.	Differentiate one way slab and two way slab with neat sketch.	3	K2
e.	Sketch parts of retaining wall and load acting on it.	5	K2
f.	Explain bi-axial bending in column.	4	K3
g.	Define-effective span, effective depth, nominal cover, location of shear reinforcement in rcc beam.	2	K2

SECTION B

2. Attempt any three of the following:

07 x 3 = 21

a.	Design a simply supported roof slab for a room 7m * 3m clear in size, imposed load 4KN/Sqm, show detailing.	3	K3
b.	What are the assumptions for the design of a reinforced concrete section for limit state of collapse in bending? Derive the stress block parameters for a rectangular cross-section.	2	K3
c.	Show that the limiting depth of neutral axis for a rectangular cross-section reinforced with Fe 415 grade steel is 0.48d.	2	K3
d.	How you will design a rcc column for axial load and bending moment combined	4	K4
e.	Sketch various types of foundation with detailing.	5	K4

SECTION C

3. Attempt any one part of the following:

07 x 1 = 07

a.	Differentiate working stress method with limit state method.	1	K2
b.	Explain stress strain diagram of Fe415.	1	K2

4. Attempt any one part of the following:

07 x 1 = 07

a.	Calculate development length for M20 and Fe 415, where you will check development length in rcc beam supporting dog-legged staircase.	2	K2
b.	Design a rectangular rcc beam for 5m effective span. Imposed load is 70 KN/m, use M20 and Fe 415.	2	K4

5. Attempt any one part of the following:

07 x 1 = 07

a.	Design a two way slab for a room 5m*6m clear in size, imposed load 4 KN/Sqm, corners not held down.	3	K4
b.	Discuss various staircases with neat sketches.	3	K2



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6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Design a short rcc column, square cross-section to carry a centroidal axial load of 1500KN, using M20 and Fe415.	4	K4
b.	Discuss IS 13920 2016 for rcc column seismic detailing.	4	K3

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Design a square spread footing to carry a column load of 1200 KN from a 350mm square tied column containing 16mm bars as longitudinal reinforcement.SBC of soil 110 KN/Sqm.consider base of footing 0.9m below the ground level. The unit wt of earth is 19.5 KN/Cum.show detailing.	5	K5
b.	Discuss structural behavior of t shaped rcc cantilever retaining wall, how you will design stem of retaining wall of 4m overall height.	5	K3